Part A Chapter 4

Neural Mechanisms for Learning: Conditioning by *Aplysia*[1]

1 Introduction

Many organisms can learn: based on certain experiences, the behaviour can change forever, or at least for longer lasting periods of time; for example, see (Gleitman, 2004, pp. 122-162). Other terms sometimes used for specific types of learning behaviour are *conditioning* or *adaptive behaviour*. For not too complex organisms, underlying neural mechanisms of learning based on long term changes in the connections between neurons (synapses) are relatively well understood; see, for example, (Gleitman, pp. 70-76, pp. 154-156; Widmaier, Raff, and Strang, 2006, pp. 176-190; Hawkins and Kandel, 1984). In particular, *Aplysia* is an appropriate species studied in this chapter, since its neural mechanisms are not too complex and have been well-investigated; *Aplysia* is a sea hare that is often used to do experiments.

1.1 Domain description and focus

Aplysia is able to learn based on the (co)occurrence of certain stimuli; for example; see also (Gleitman, 2004, pp. 154-156). The behaviour before a learning phase is as follows:

- a tail shock leads to a response (contraction)
- a light touch on its siphon is insufficient to trigger such a response

Here the siphon is a tube-shaped organ from which water is released. Now suppose that in a training period the following experimental protocol is undertaken. In each trial the subject is

- touched lightly on its siphon and then immediately,
- shocked on its tail, and
- as a consequence it responds by a contraction.

After a number of trials (four in the current example) the behaviour has changed:

- the animal also responds (contracts) to a siphon touch.

[1] Based on: Bosse, T., Jonker, C.M., and Treur, J., (2007). Simulation and Analysis of Adaptive Agents: an Integrative Modelling Approach. *Advances in Complex Systems Journal*, vol. 10, 2007, pp. 335 - 357.

Roughly spoken the internal neural mechanism for Aplysia's conditioning can be depicted as in Figure 1, following (Gleitman, 2004, pp. 154-156).

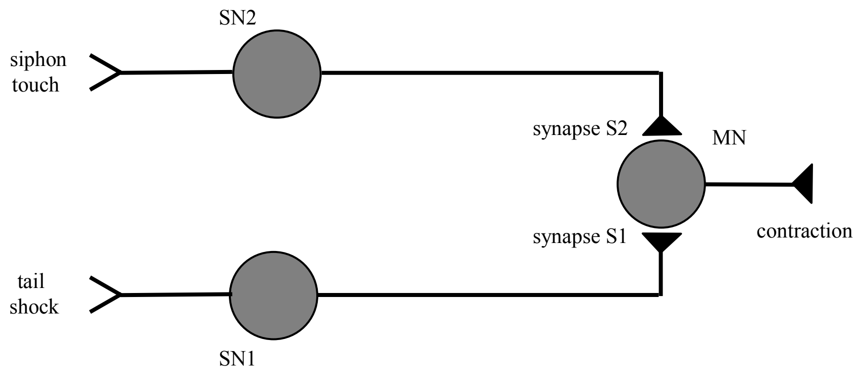

Figure 1 Neural mechanism for conditioning in *Aplysia*

A tail shock activates a sensory neuron SN1. Activation of this neuron SN1 activates the motoneuron MN via the synapse between the two neurons; activation of MN makes the sea hare move. A siphon touch activates the sensory neuron SN2. Activation of this sensory neuron SN2 normally does not have sufficient impact on MN to activate MN, as the synapse between SN2 and MN is not strong enough. After learning, the synapse has become stronger and therefore activation of SN2 has sufficient impact to activate MN. If both SN2 and MN are activated simultaneously, this changes the strength of the synapse between SN2 and MN: the effect is that in this synapse more neurotransmitter is produced whenever SN2 is activated. After a number of times this leads to the situation that this synapse is strong enough so that also activation of SN2 leads to activation of MN.

1.2 Characteristic patterns of Aplysia behaviour

In this chapter a model of *Aplysia*'s neural mechanism for conditioning will be discussed that should exhibit following patterns of behaviour:

- Initially a contraction occurs after a shock, but no reaction upon a siphon touch.
- After a training period of trials (each of a siphon touch followed by a tail shock), contraction will take place after any siphon touch.
- During a training period the strength of the synapse increases and eventually reaches a maximal strength.

2 Analysis of the Main Aspects and Their Relations

In this section the main concepts and their relations needed for the model are identified.

2.1 Identification of relevant concepts

Following the description in Section 1, external stimuli are used, and an action: *siphon touch*, *tail shock, contraction*. Furthermore *activations of sensory n*eurons SN1 and SN2 and *motoneuron* MN play a role. Finally, the *strength of the synapses* (S1 between SN1 and MN and S2 between SN2 and MN) are crucial for the process. In summary, the following concepts are used:

- tail shock
- siphon touch
- contraction
- sensory neuron SN1 is activated
- sensory neuron SN2 is activated
- motoneuron MN is activated
- the synapse S1 between SN1 and MN has a certain strength
- the synapse S2 between SN2 and MN has a certain strength

2.2 Relationships between the identified concepts

The relationships between the identified concepts are as follows (see also Figure 1).

Sensory processes

- A tail shock affects activation of SN1
- A siphon touch affects activation of SN2

Action generation

- Activation of SN1 affects activation of MN
- Activation of SN2 together with a certain strength of synapse S2 affects activation of MN
- Activation of MN affects contraction

Adaptation

- Simultaneous activation of MN and SN2 affects the strength of synapse S2 between SN2 and MN by increasing it

In graphical form these relationships are depicted in Figure 2. Note that, in contrast to Figure 1, here the motor neuron MN and its connections are described by three concepts, one for its activation state, and two for the two states of each of the synapses: S1 between SN1 and MN and S2 between SN2 and MN. Notice that, the strength of the former synapse is assumed constant, and is therefore not included in the relationships in the model.

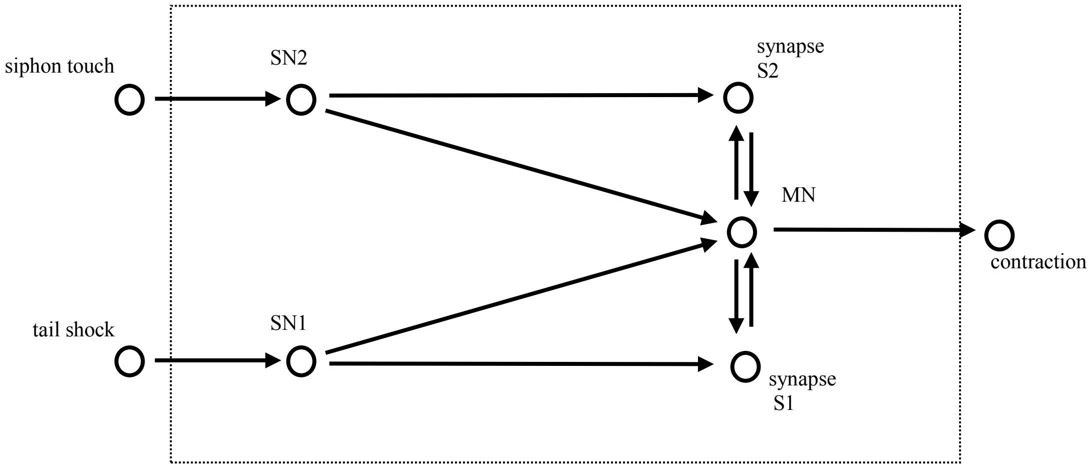

Figure 2 Dynamic relationships in the model

3 The Detailed Model

In this section the detailed model is introduced. It is a variation of the model described in (Bosse, Jonker, and Treur, 2007). First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a numerical variable is introduced that takes natural numbers as values; see Table 1.

stimuli	
tail shock	*TailShock*
siphon touch	*SiphonTouch*
sensory representations	
sensory neuron SN1 is activated	*ASN1*
sensory neuron SN2 is activated	*ASN2*
adaptation states	
the synapse strength for the connection SN1/MN	*S1*
the synapse strength for the connection SN2/MN	*S2*
preparation state	
motoneuron MN is activated	*AMN*
action	
contraction	*Contraction*

Table 1 Formalisation of the relevant conc epts by numerical variables

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships, which are analysed in more detail below. They describe the basic parts of the process; see also Figure 3.

Sensory processes

From tail shock to activation of SN1

If a tail shock occurs, then sensory neuron SN1 will be activated for a time duration d (for convenience taken three times Δt).

$$ASN1(t+\Delta t) = max(TailShock(t), TailShock(t-\Delta t), TailShock(t-2\Delta t))$$

From siphon touch to activation of SN2

If a siphon touch occurs, then sensory neuron SN2 will be activated for a time duration d (for convenience taken three times Δt).

$$ASN2(t+\Delta t) = max(SiphonTouch(t), SiphonTouch(t-\Delta t), SiphonTouch(t-2\Delta t))$$

Action generation

From activation of SN1 or SN2 to activation of MN

If activation of SN1occurs, or activation of SN2 occurs and the synapse has strenght 4, then MN will be activated.

$$AMN(t+\Delta t) = \begin{cases} 1 & \text{if } ASN1(t) = 1 \text{ or: } S(t) = 4 \text{ and } ASN2(t) = 1 \\ 0 & \text{else} \end{cases}$$

From activation of MN to contraction

If activation of MN occurs, then contraction will take place

$$Contraction(t+\Delta t) = AMN(t)$$

Adaptation

Adjustment of the synapse strength based on simultaneous activation of MN and SN2

If activation of SN2 occurs, and activation of MN occurs, and the synapse S2 between SN2 and MN has strenght r with r < 4, then this synapse will have strenght r+1.

$$S2(t+\Delta t) = \begin{cases} S2(t) + 1 & \text{if } S2(t) < 4 \text{ and } AMN(t) = ASN2(t) = 1 \\ S2(t) & \text{else} \end{cases}$$

In Table 2 the dependencies over time are shown based on the above temporal relationships.

t	tail shock	siphon touch	ASN1	ASN2	AMN	synapse S2	contrac-tion
0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0
2	0	1	1	0	0	0	0
3	0	0	1	1	0	0	0

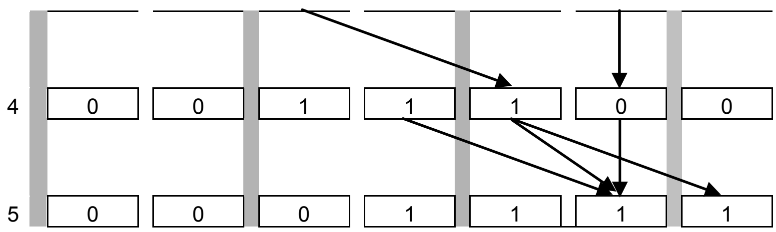

Table 2 Dependencies between variables over time

4 Simulation Experiments

Simulations have been performed in Microsoft Excel. One example trace is shown in Figure 4. As can be seen in Figure 4, at the beginning of the trace the organism has not performed any conditioning. The initial siphon touch it receives at time point 1 does lead to the activation of sensory neuron SN2, but the synapse between SN2 and motoneuron MN does not produce much neurotransmitter yet (indicated by *S2* being *0*). Thus, the activation of SN2 does not yield an activation of MN, and consequently no external action follows.

In contrast, it is shown that a shock of the organism's tail does initially lead to the external action of contraction. This can be seen in Figure 4 between time point 8 (when the tail shock occurs) and time point 10 (when the animal contracts). This is part of the first trial of the learning phase. This phase consists of a sequence of four trials where a siphon touch is immediately followed by a tail shock.

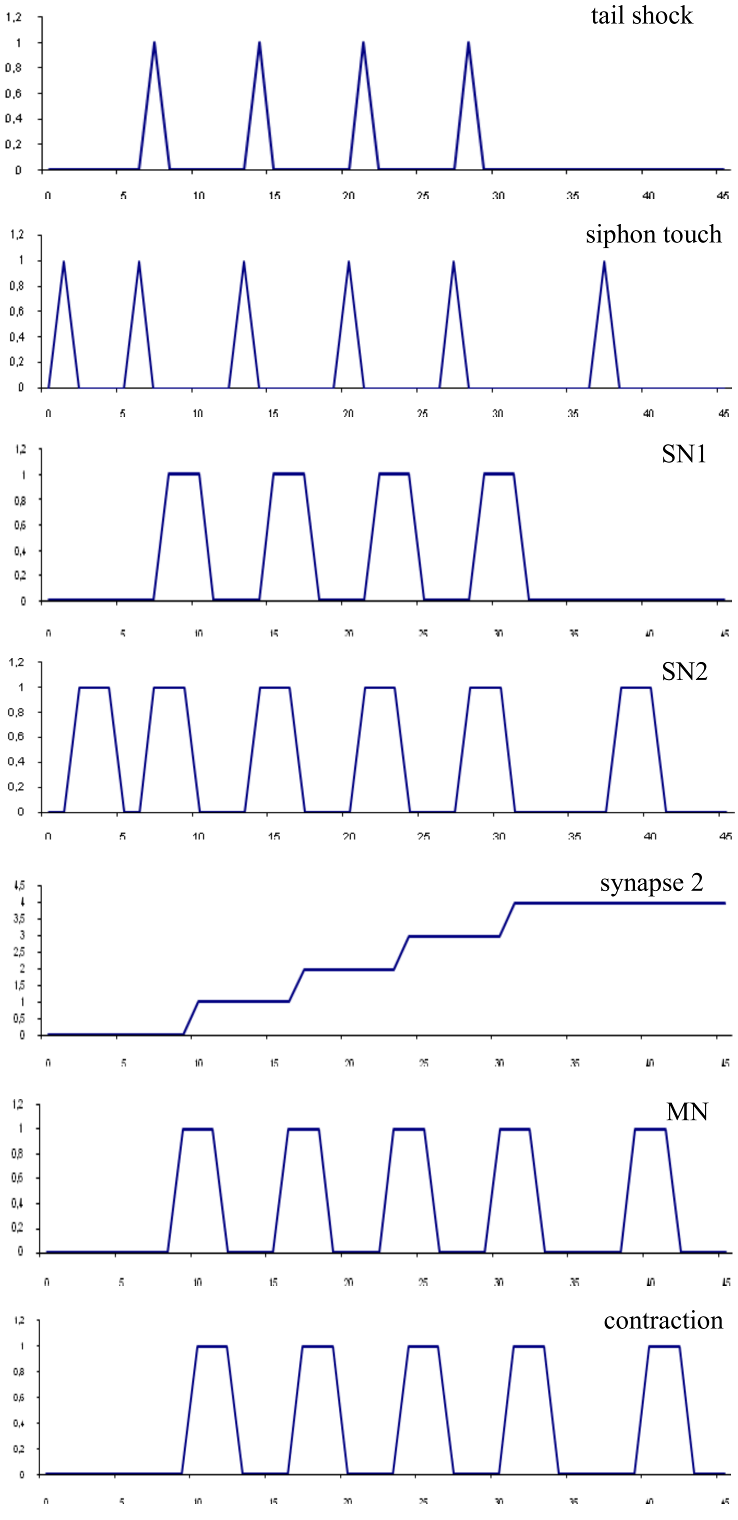

Figure 4 Example simulation trace of *Aplysia*'s conditioning

As a result, the sensory neuron SN2 is activated at the same time as the intermediary neuron MN, which causes the synapse to produce an increased amount of neurotransmitter upon activation. Such an increment of the amount of neurotransmitter during activations is indicated by a transition of *S2*: first from *1* to *2*, then to *3*, and finally to *4*. As soon as it is *4* (at time point 31), the conditioning process has been performed successfully. From that moment, Aplysia's behaviour has changed: it also contracts on a siphon touch, as shows at the end of the period.

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally some possibilities of modification or refinement of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

A number of different simulations conducted in Excel show the characteristic patterns identified in Section 1.2. It is easy to perform a few more experiments to check more training scenarios.

5.2 Assumptions behind the model

The following assumptions were made.

- The adjustments of the synapse strength are linear: every trial the same increase, until a maximum is reached.
- There is no decay of synapse strength.
- Activations of neurons last three time steps.
- In a training period the stimuli (siphon touch and tail shock) are presented at two successive time steps.

5.3 Possible variations and refinements

It is not too difficult to refine the model into a more complex model to relax the assumptions that were made. For example:

- Faster or slower learning: complete training periods of different lengths.
- The adjustments of the synapse strength can be taken not linear but bounded by a maximum.
- Activation of the motoneuron MN by sensory neuron SN2 can take place in case the strength of the synapse connecting them is above a certain threshold, not at a maximum value.
- Decay of synapse strength can be incorporated.
- The activation periods of neurons may be taken to last longer or shorter than three time steps.
- In a training period the time difference between the two stimuli (tail shock and siphon touch) can be varied.
- Between stimulus and MN not one but a number of neurons can be used, both serial and in parallel, that together (in sum) realise activation of MN.

6 Further Explorations

A large body of literature is available reporting how different types of animals show adaptive behaviour in laboratory environments; e.g., (Skinner, 1953).

Sensory Preconditioning
A typical setting for rats is the situation that a rat gets light foot shocks on which it reacts by leg flexions. This is often explained by assuming the neural connections shown in Figure 5.

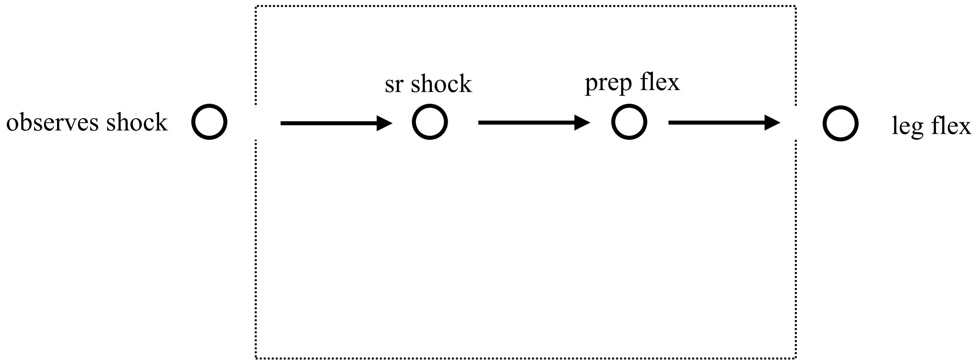

Figure 5 Neural path from shock to leg flexion

Here the first neural state property is an internal sensory representation of the shock, and the second an internal preparation for the action leg flexion. Upon hearing a tone, usually no leg flexion is shown. If during such experiments a number of times first a tone is heard by the animal, after which a foot shock occurs, the behaviour becomes different. After some time the animal shows leg flexions upon hearing the tone, without a foot shock being present. This can be described by assuming that due to learning an additional neural pathway has been achieved, as shown in Figure 6.

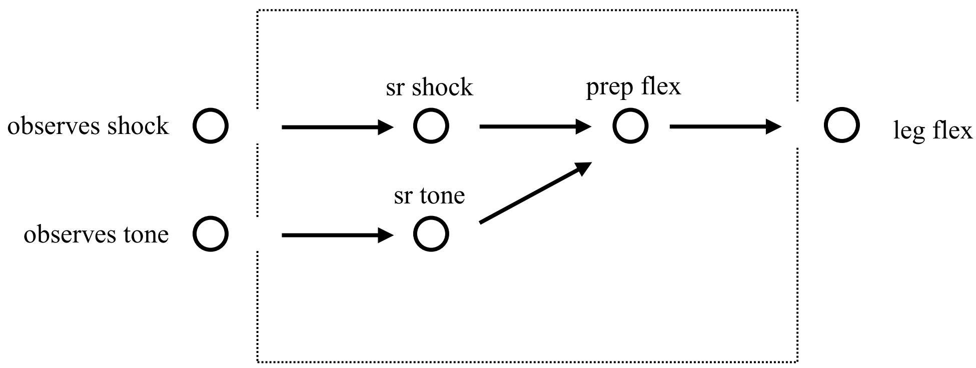

Figure 6 Path from sensory representation of the tone to preparation of leg flexion

However, also an alternative pathway is possible that explains the same behaviour, but which does not assume a direct connection from the sensory representation of the tone to the preparation of the flexion, but rather from the sensory representation of the tone to the sensory representation of the shock, as shown in Figure 7.

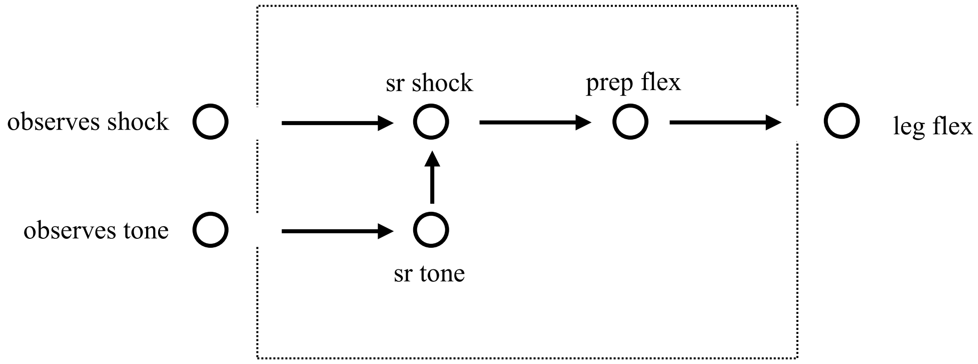

Figure 7 Path from sensory representation of the tone to sensory representation of the shock

This neural pathway describes what is sometimes called *conditioned perception* or *sensory preconditioning*.

A decisive experiment

To find out whether sensory preconditioning occurs, an experiment has been conducted that runs as follows; see (Brogden, 1939).

I. Present the animal with repeated paired tone and light stimuli. No overt response.
II. Pair the tone with a foot shock until the tone elicits a conditioned flexion response
III. Present the light alone.

The outcomes of these experiments make clear that sensory preconditioning indeed takes place: the animal shows flexion behaviour in response to the light. A variant of the second picture above (in Figure 7) can account for these outcomes: a pathway is created from the sensory representation of the light to the sensory representation of the tone during phase I of the experiment, and one from the sensory representation of the tone to the sensory representation of the food shock in phase II; see Figure 8.

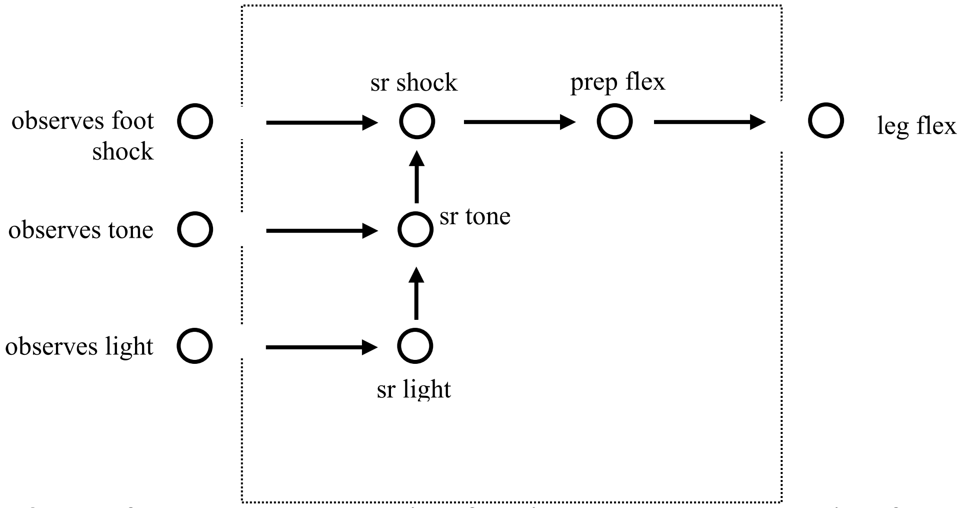

Figure 8 Paths from sensory representation of the light to sensory representation of the tone and from sensory representation of the tone to sensory representation of the shock

These descriptions only show the internal dynamics behind the behaviours before and after adaptation. The process of adaptive behaviour itself is not depicted in these diagrams, and therefore is not described in these diagrams. The adaptive process itself can be described in the following manner (similar to the model for *Aplysia*'s conditioning):

- For each connection that is learned, add a state property describing the strength of the connection, for example from *0* to *4*, with *0* the initial strength
- Specify executable dynamic relationships describing how these strengths are changed during the learning process; for example, increased by one after each trial, until *4* is reached,
- Specify executable dynamic relationships describing how a connection of a certain strength affects the further dynamics; for example, the effect only occurs if the strength is *4*.

References

Bosse, T., Jonker, C.M., and Treur, J., (2007). Simulation and Analysis of Adaptive Agents: an Integrative Modelling Approach. *Advances in Complex Systems Journal*, vol. 10, 2007, in press. Shorter version in: *Proc. of the 39th Annual Simulation Symposium.* IEEE Computer Society Press, 2006, pp. 312-319.

Brogden, W. J. (1939). Sensory preconditioning. *Journal of Experimental Psychology* **25**, 323-332.

Gleitman, H. (1999, 2004). *Psychology*, W.W. Norton & Company, New York.

Hawkins, R.D., and Kandel, E.R. (1984). Is There a Cell-Biological Alphabet for Simple Forms of Learning? *Psychological Review*, vol. 91, 1984, pp. 375-391.

Hesslow, G. (2002) Conscious thought as simulation of behaviour and perception. *Trends in Cognitive Sciences*, vol. 6, pp. 242-247

Skinner, B. F. (1953). *Science and Human behavior* Macmillan, New York

Widmaier, E.P., Raff, H., and Strang, K.T. (2006). *Vander's Human Physiology: the Mechanisms of Body Functioning*, MacGraw Hill, 10th edition.